Syed Mohammad Huzaifa

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Education

NED University of Engineering & Technology Bachelor of Science in Computer Science, AI Specialization (BS-CS), Karachi, Pakistan, Sep 2023 to Sep 2027

Aga Khan Higher Secondary School (AKHSS) Higher Secondary School Certificate (HSSC), Pre-Engineering, Karachi, Pakistan, 2021 to 2023

SMS Aga Khan School, Karimabad Secondary School Certificate (SSC), Karachi, Pakistan, 2019 to 2021

Experience

FlyRank, ML Intern Jul 2026 to Sep 2026

- Built and evaluated 4 classifiers (logistic regression, random forest, decision tree, gradient boosting) ranking pages by under-capture risk over a 79M-row anonymized dataset, queried with DuckDB, using GroupKFold split by client to prevent train/test leakage.

- Matched but did not beat a frozen baseline rule on precision@50 (0.92 both, against a 0.254 base rate). Models added reach, flagging 44 of 6,053 low-volume pages the rule missed, and shipped without a value estimate since 42% of flagged pages self-resolve within 30 days untouched.

Technical Skills

ML / AI: PyTorch, scikit-learn (Pipelines, GroupKFold), Pandas, NumPy, DuckDB, Linear and Logistic Regression, KNN, Random Forest, XGBoost, Gradient Boosting, Decision Trees, formal verification (SHERLOCK, MILP with PuLP), pdfplumber and regex-based extraction, Data Preprocessing, Model Evaluation

Algorithms: Merge Sort, Quick Sort, heap-based priority queues, Big-O complexity analysis, benchmarking

Programming Languages: Python (primary, used across every project below), C/C++, PHP, JavaScript, SQL

Web Development & Databases: HTML, CSS, Bootstrap, ASP.NET Core, SQL Server

Tools & Platforms: Git, GitHub Actions (CI), Jupyter, Google Colab, WordPress, Microsoft Office Suite

Soft Skills: Analytical Thinking, Problem Solving, Team Collaboration, Fast Learner, English and Urdu

Projects

Supreme Court of Pakistan Judgment Analytics Python, pandas, scikit-learn, pdfplumber, 2026

- Built a pipeline extracting structured data (25 fields) from about 1,200 unstructured Pakistani Supreme Court judgment PDFs, using pdfplumber and hand-written regex.
- Deduplicated cases across judges and benches, then evaluated Random Forest, XGBoost, and a linear SVM against a majority-class baseline rather than raw accuracy.

Neural Network Output Verification, Air Quality Data PyTorch, PuLP/CBC (MILP), 2026

- Implemented the SHERLOCK algorithm to formally verify output bounds of a ReLU feedforward network (PyTorch) predicting CO concentration, combining local search with an exact MILP solve.
- Trained the network to R-squared 0.93 and produced certified output intervals instead of point predictions, the worst case a warning system could report.

Social Media Feed Ranking Benchmark Python, heapq, matplotlib, 2026

- Implemented and benchmarked Merge Sort, Quick Sort, Insertion Sort, and a heap-based priority queue ranking posts by engagement score across dataset sizes from 100 to 100,000.
- Measured heap-based incremental updates at roughly 22x faster than a full re-sort when inserting a new post into an already-ranked 10,000-post feed.

Certifications

- Build an AI Personal Assistant with a Vector Database, Scrimba
- Python Project for Data Science, IBM
- Python for Data Science, AI & Development, IBM